Chapter 5, Section 4

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1. The linear system of conics in $\mathbf{P}^2$ with two assigned base points P_1 and P_2 (4.1) determines a morphism ψ of X' (which is $\mathbf{P}^2$ with P_1 and P_2 blown up) to a nonsingular quadric surface Y in $\mathbf{P}^3$, and furthermore X' via ψ is isomorphic to Y with one point blown up.

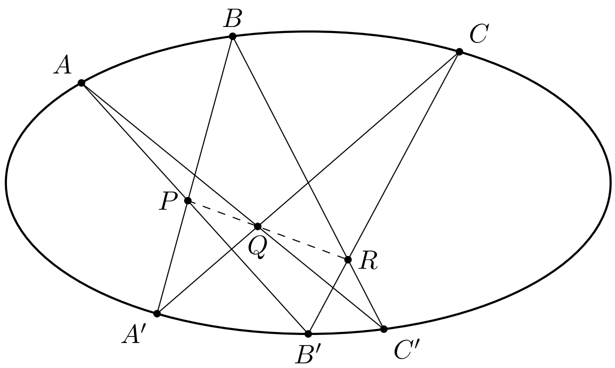
2. Let φ be the quadratic transformation of (4.2.3), centered at P_1, P_2, P_3. If C is an irreducible curve of degree d in $\mathbf{P}^2$, with points of multiplicity r_1, r_2, r_3 at P_1, P_2, P_3, then the strict transform C' of C by φ has degree $d' = 2d - r_1 - r_2 - r_3$, and has points of multiplicity $d - r_2 - r_3$ at Q_1, $d - r_1 - r_3$ at Q_2, and $d - r_1 - r_2$ at Q_3. The curve may have arbitrary singularities.

3. Let C be an irreducible curve in $\mathbf{P}^2$. Then there exists a finite sequence of quadratic transformations, centered at suitable triples of points, so that the strict transform C' of C has only *ordinary* singularities, i.e., multiple points with all distinct tangent directions.

4. (a) Use (4.5) to prove the following lemma on cubics: If C is an irreducible plane cubic curve, if L is a line meeting C in points P, Q, R, and L' is a line meeting C in points $P', Q'R'$, let P' be the third intersection of the line PP' with C, and define Q'', R'' similarly. Then P'', Q'', R'' are collinear.

(b) Let P_0 be an inflection point of C, and *deifne* the group operation on the set of regular points of C by the geometric recipe "let the line PQ meet C at R, and let P_0R meet C at T, then $P + Q = T$" as in (II, 6.10.2) and (II, 6.11.4). Use (a) to show that this operation is associative.

5. Prove Pascal's theorem: if A, B, C, A', B', C' are any six points on a conic, then the points $P = AB'.A'B$, $Q = AC'.A'C$, and $R = BC'.B'C$ are collinear.



6. Generalize (4.5) as follows: given 13 points $P_1, \ldots, P_{13}$ in the plane, there are three additional deterined points P_{14}, P_{15}, P_{16}, such that all quartic curves through $P_1, \ldots, P_{13}$ also pass through P_{14}, P_{15}, P_{16}. What hypotheses are necessary on $P_1, \ldots, P_{13}$ for this to be true.

7. If D is any divisor of degree d on the cubic surface (4.7.3), show that

$$p_a(D) \leq \begin{cases} \frac{1}{6}(d-1)(d-2) & \text{if } d \equiv 1, 2 \mod 3 \\ \frac{1}{6}(d-1)(d-2) + \frac{2}{3} & \text{if } d \equiv 0 \mod 3. \end{cases}$$

Show furthermore that for every $d > 0$, this maximum is achieved by some irreducible nonsingular curve.

8. Show that a divisor class D on the cubic surface contains an irreducible curve $\iff$ it contains an irreducible nonsingular curve $\iff$ it is either (a) one of the 27 lines, or (b) a conic (meaning a curve of degree 2) with $D^2 = 0$, or (c) $D.L \geq 0$ for every line L, and $D^2 > 0$.

9. If C is an irreducible non-singular curve of degree d on the cubic surface, and if the genus $g > 0$, then

$$g \geq \begin{cases} \frac{1}{2}(d-6) & \text{if } d \text{ is even, } d \geq 8, \\ \frac{1}{2}(d-5) & \text{if } d \text{ is odd, } d \geq 13, \end{cases}$$

and this minimum value of $g > 0$ is achieved for each d in the range given.

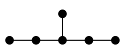
10. A curious consequence of the implications (iv) $\implies$ (iii) of (4.11) is the following numerical fact: Given integers $a, b_1, \dots, b_6$ such that $b_i > 0$ for each i, $a - b_i - b_j > 0$ for each i, j, and $2a - \sum_{i \neq j} b_i > 0$ for each j, we must necessarily have $a^2 - \sum b_i^2 > 0$. Prove this directly (for $a, b_1, \dots, b_6 \in \mathbf{R}$) using methods of freshman calculus.

11. *The Weyl Groups.* Given any diagram consisting of points and line segments joining some of them, we define an abstract group, given by the generators and relations, as follows: each point represents a generator x_i. The relations are $x_i^2 = 1$ for each i; $(x_i x_j)^2 = 1$ if i and j are not joined by a line segment, and $(x_i x_j)^3 = 1$ if i and j are joined by a line segment.

(a) The *Weyl group* $\mathbf{A}_n$ is defined using the diagram

of $n-1$ points, each joined to the next. Show that it is isomorphic to the symmetric group Σ_n as follows: map the generators of $\mathbf{A}_n$ to the elements $(12), (23), \dots, (n-1, n)$ of Σ_n, to get a surjective homomorphism $\mathbf{A}_n \to \Sigma_n$. Then estimate the number of elements of $\mathbf{A}_n$ to show in fact it is an isomorphism.

(b) The *Weyl group* $\mathbf{E}_6$ is defined using the diagram



Call the generators $x_1, \dots, x_5$ and y. Show that one obtains a surjective homomorphism $\mathbf{E}_6 \to G$, the group of automorphisms of the configuration of 27 lines (4.10.1), by sending $x_1, \dots, x_5$ to the permutations $(12), (23), \dots, (56)$ of the E_i, respectively, and y to the element associated with the quadratic transformation based at P_1, P_2, P_3.

(c) Estimate the number of elements in $\mathbf{E}_6$, and thus conclude that $\mathbf{E}_6 \cong G$.

12. Use (4.11) to show that if D is any ample divisor on the cubic surface X, then $\mathrm{H}^1(X, \mathcal{O}_X(-D)) = 0$. This is Kodaira's vanishing theorem for the cubic surface (III, 7.15).

13. Let X be the Del Pezzo surface of degree 4 in $\mathbf{P}^4$ obtained by blowing up 5 points of $\mathbf{P}^2$ (4.7).

(a) Show that X contains 16 lines.

(b) Show that X is a complete intersection of two quadric hypersurfaces in $\mathbf{P}^4$ (the converse follows from (4.7.1)).

14. Using the method of (4.13.1), verify that there are nonsingular curves in $\mathbf{P}^3$ with $d = 8$, $g = 6, 7$; $d = 9$, $g = 7, 8, 9$; $d = 10$, $g = 8, 9, 10, 11$. Combining with (IV, §6), this completes the determination of all possible g for curves of degree $d \geq 10$ in $\mathbf{P}^3$.

15. Let $P_1, \dots, P_4$ be a fintie set of (ordinary) points of $\mathbf{P}^2$, no 3 collinear. We define an *admissible transformation* to be a quadratic transformation (4.2.3) centered at some three of the P_i (call them P_1, P_2, P_3). This gives a new $\mathbf{P}^2$, and a new set of r points, namely Q_1, Q_2, Q_3, and the images of $P_4, \dots, P_r$. We say that $P_1, \dots, P_r$ are in *general position* if no three are collinear, and furthermore after any finite sequence of admissible transformations, the new set of r points also have no three collinear.

(a) A set of 6 points is in general position if and only if no three are collinear and not all six lie on a conic.

(b) If $P_1, \dots, P_r$ are in general position, then the r points obtained by any finite sequence of admissible transformations are also in general position.

(c) Assume the ground field k is uncountable. Then given $P_1, \dots, P_r$ in general position, there is a dense $V \subseteq \mathbf{P}^2$ such that for any $P_{r+1} \in V$, $P_1, \dots, P_{r+1}$ will be in general position.

(d) Now take $P_1, \dots, P_r \in \mathbf{P}^2$ in general position, and let X be the surface obtained by blowing up $P_1, \dots, P_r$. If $r = 7$, show that X has exactly 56 irreducible nonsingular curves C with $g = 0$, $C^2 = -1$, and that these are the only irreducible curves with negative self-intersection. Ditto for $r = 8$, the number being 240.

(e) For $r = 9$, show that the surface X defined in (d) has infinitely many irreducible nonsingular curves C with $g = 0$ and $C^2 = -1$.

16. For the *Fermat cubic surface.* $x_0^3 + x_1^3 + x_2^3 + x_3^3 = 0$, find the equations of the 27 lines explicitly, and verify their incidence relations. What is the group of automorphisms of this surface.